MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Business Technology Proposal for MotorPH Payroll System

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<**IMPORTANT NOTE:** The Executive Summary is intended to occupy an entire page. Transfer the succeeding contents to the next page of the document. **Delete this note before submission**>

Executive Summary

<The executive summary provides a high-level overview of the proposal's main points. It is meant to give readers a quick overview of the contents without needing them to review the entire document.

Structure the executive summary by writing it based on the outline below:

- *Problem Statement - Introduce the proposal's main problem or opportunity in a few sentences. Highlight the importance of this problem in the context of business operations and goals.*
- *Proposed Solution - Summarize the proposed technological solution to the problem by emphasizing the system's primary features and functions.*
- *Benefits and Value Proposition - Outline the major advantages that the technological solution brings to the business. Highlight how the solution corresponds with the company's goals and strategy.*
- *Scope and Implementation - Present how and where the technology will be employed. Mention any specific phases or milestones in the setup process if needed.*
- *Stakeholders and Collaboration - Mention the project's internal and external stakeholders. Highlight the proposal's collaborative nature and how it will include multiple departments or teams.*
- *Risk Awareness - Recognize any dangers or problems related to the proposed technology solution. Mention risk-mitigation techniques to be put in place.*
- *Next Steps - Include a future statement on the proposal process. Include details on possibilities of further discussions, approvals, or actions.*
- *Call to Action - Encourage readers to seriously examine the proposal and take the required steps to move forward. Provide contact details in case of additional questions or conversations.>*

1. Introduction

MotorPH has long relied on a patchwork of paper timecards, spreadsheet-driven calculations, and department-specific record alignment, with each minor amendment rippling into fresh delays, occasional miscalculations, and growing exasperation. Conceivably, what may have begun as a straightforward setup has evolved into a repetitive cycle of data entry and tedious approvals, resulting in late payslips and bottlenecked workflows. By contrast, the MotorPH Payroll System that we're proposing aspires to diverge from the series-based processes and drive the operations between departments into a parallel and modular system that refashion attendance logging, salary computation, leave management and statutory contributions within a unified, automated process; and thus, thereby cutting turnaround times, enhancing precision and freeing staff to focus on more meaningful tasks.

1.1. Business Context

As MotorPH's business operations gradually spread their branches across the nation, prevailing trends in HR and payroll management suggest a shift toward complete system autonomy, cloud-ready solutions that offer real-time visibility and effortless scalability. At the same time, the proliferation of hybrid and remote work models calls for mobile-friendly time-keeping and reliably secured off-site attendance tracking, simultaneously tightening data-privacy standards, magnifying the need to protect SSS, TIN, and PhilHealth identifiers. Although MotorPH is yet to operate on a cloud-hosted platform, adopting such a system in the future could deliver encrypted storage, role-based access controls, and detailed audit trails, capabilities that legacy paper methods simply cannot match; legacy paper methods risk breaches, whereas modern cloud platforms offer encrypted storage and fine-grained access controls for PII.

Whilst these auxiliary features were not the immediate focus of our initial assessment, they level out payroll facilitation, which remains integral to disentangling the once messy dispute resolution process and ensuring unlined end-to-end processing. Accordingly, the system incorporates a dedicated Payroll Supervision module that handles dispute tickets, applies individual, team, or global adjustments, and enforces approval workflows with full audit logging. It provides consolidated payroll reporting and automated payslip distribution (complete with retry logic for failed email deliveries) so that every stakeholder, from payroll staff to line managers, can track changes, review historic runs, and receive final payslips without delay.

1.2. Challenges and Opportunities

MotorPH currently contends with a recalcitrant assortment of manual routines that impede efficiency, yet within those very impediments are the derivative avenues that form the basis for enhancing the MotorPH workflow.

MotorPH Challenges:

- Fragmented manual processes: Myopically department-centred and isolated procedures, jeopardising cross-functional cohesion, and rendering it ineffective and non-viable in the long-term: reliance on paper timecards and spreadsheet-based timesheets has fostered a mire of calculation errors, delayed payslips, and recurrent payroll disputes, eroding both morale and trust.
- Security gaps: In making a centralised portal for dashboards, the absence of rigorous two-factor authentication and fine-grained access controls leaves sensitive PII (SSS, TIN, PhilHealth data) vulnerable to data breaches and incursions.
- Compliance burden: Manually compiling statutory contributions for SSS, PhilHealth, Pag-IBIG, and taxes alongside the already arduous task of manual payroll management is laborious and error-prone, risking costly regulatory sanctions.
- Lack of integration: Without a centralised data hub, branches must compare differing records through a tangled mess of manual communication, an exercise in redundancy that precipitates miscommunication and inefficiency.

System Opportunities:

- Modular automated workflows: An integrated payroll platform can diminish, if not obliterate, repetitive data entry, enabling end-to-end automation of attendance, salary computation, and leave management, and thus, slashing processing times and minimising errors.
- Centralised, real-time data: A cloud-ready solution promises up-to-the-minute visibility into employee and payroll metrics across all involved departments, elevating coordination and data-driven decision-making.
- Employee self-service: With the accessibility of the MotorPH dashboard available also as a mobile app, it allows staff to do things off-site that would otherwise need in-office presence (clock in/out, submit leave

requests, and view payslip projections autonomously), supporting a hybrid and remote-capable workforce.

- Enhanced compliance and audit: Built-in audit trails, comprehensive logs, and automated statutory reporting simplify inspections and dispute resolution, ensuring transparent management of every payroll action.

1.3. Significance of the Proposal

Lacking sufficient modernisation, MotorPH stands to become a victim of operational drag when staffing and nationwide branches increase, to the detriment of efficiency and staff morale, with continued manual, siloed processes, distracting resources from strategic growth initiatives. Alternatively, by adopting a modular, service-oriented payroll architecture, MotorPH can unleash quicker processing, more transparent insight, and a more vital support model, something that not only protects day-to-day and repetitive operations but enables the business to leverage future innovation: whether next-generation analytics, AI-driven forecasting, or effortless integration with other enterprise systems.

1.4. Objectives of the Proposal

Each objective directly addresses identified challenges and harnesses the opportunities at hand, with clear metrics for success:

- Automate more than fifty per cent of manual payroll tasks within months of integration, by deploying modular services for time capture, salary computation, and leave adjudication—thereby reducing error-laden entries and dispute volume.
- Integrate all five branches into a single data platform as soon as possible, ensuring real-time synchronisation of attendance and payroll records and eliminating department-specific process isolation.
- Achieve high employee adoption of self-service features within months of launch, measured by mobile app utilisation rates for clock-in/out, salary snapshot reviews, and ticket submissions.
- Resolve most, if not all, of payroll disputes within 48 hours, through the new ticketing and audit-trail capabilities, cutting average resolution time and bolstering staff confidence.
- Maintain around-the-clock system availability, leveraging possible scalable cloud infrastructure and modular service redundancy to ensure uninterrupted access to payroll and attendance functions.

2. Technology Solution

<This section contains a full description of the proposed technology, including its features and functions and how it immediately contributes to solving the challenges or capitalizing on the possibilities stated previously in the proposal. Begin by presenting a high-level overview of the technology solution and then succeed by providing inputs on the subsections provided below>

2.1. Solution Overview

<Introduce the suggested technical solution briefly. Give a high-level overview of the solution's main objective and how it provides opportunities for MotorPH or solves business challenges.>

2.2. Key Features & Functionality

- **Basic Employee Interactions Module**
  - Login: secure authentication with 2FA and session management
    - **Impact:** Fortifies entryways against unauthorised access, obviating credential-based exploits and instilling confidence that only verified personnel can engage with downstream services.
  - Clock In/Out: off-site clock-in, QR-based on-site verification, clock-out confirmation, and early-leave warning
    - **Impact:** Automates time capture, truncating manual entry errors and delivering instantaneous office presence information accessible to managers without the burdensome need of cross-checking records.
  - View Attendance: personal time-in/out logs, counts of present/absent/tardy days, overtime/undertime hours, leave usage
    - **Impact:** Equips employees with transparent, real-time visibility into their attendance records, reducing overpopulation of unresolved inquiry tickets of the same regard and plays a part in raising self-service autonomy.
  - Leave Calendar: bi-monthly team leave calendar and individual leave application status

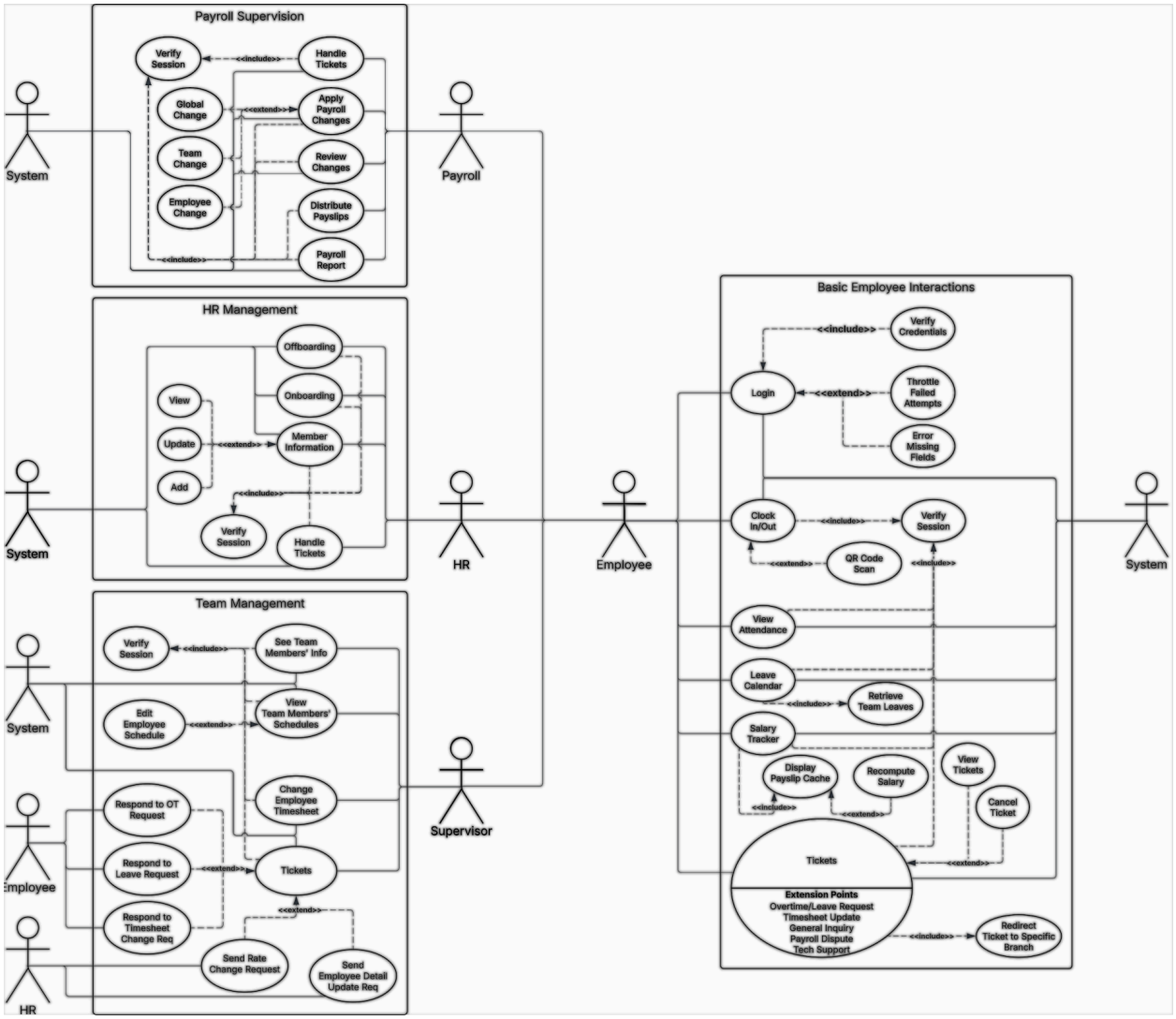
        - **Impact:** Collapses leave schedules into a single, dynamic view—preventing gaps in cover, cutting the frills in staffing plans and removing splintered spreadsheets.
    - Salary Tracker: precomputed earnings snapshot (hours, incentives, deductions, net pay) with timestamped estimates
        - **Impact:** Furnishes near-instant payroll projections, minimising uncertainty and curtailing back-and-forth queries (and tickets) related to net-pay calculations.
    - Tickets: submit and track requests (overtime, leave, timesheet updates, payroll disputes, tech support, general inquiries)
        - **Impact:** Bundles all employee requests into an auditable queue, accelerating resolution cycles and ensuring no ticket vanishes into procedural obscurity.
- **Team Management Module**
    - View Team Members' Info: roster with contact details and availability
        - **Impact:** Grants leads a consolidated snapshot of team composition and status, obviating time-consuming ad hoc checks and enabling rapid resource allocations.
    - View/Edit Schedules: single-day, multi-day, and weekly pattern shift assignments between team members
        - **Impact:** Enables quick schedule modifications through an intuitive interface, reducing shift-planning friction and preempting coverage conflicts.
    - Change Employee Timesheet: approve or edit attendance entries
        - **Impact:** Accelerates adjudication of timesheet discrepancies, slashing the lag between correction and payroll finalisation.
    - Tickets: process overtime/leave/timesheet/reschedule requests for team members
        - **Impact:** Empowers supervisors to manage subordinate requests without routing through means of face-to-face, direct contact, or HR, diminishing bureaucratic latency.
- **HR Management Module**
    - Onboarding: add new hires with required personal and tax details

        - **Impact:** Condenses the hiring lifecycle into a guided workflow, ensuring all requisite data is captured uniformly and expediting time-to-productivity.
    - Offboarding: process departures through clearance steps
        - **Impact:** Standardises exit procedures, mitigating the risk of orphaned accounts (active accounts that are otherwise inactive in reality) or lingering access privileges.
    - View/Update/Add Member Info: search and modify employee records, including mid-cycle rate changes
        - **Impact:** Facilitates swift adjustments to employee profiles, ensuring compensation revisions propagate seamlessly through payroll calculations.
    - Handle Tickets: resolve HR-direct and general inquiry tickets
        - **Impact:** Provides HR with a unified ticketing interface, consolidating disparate queries and accelerating response times.
- **Payroll Supervision Module**
    - Handle Tickets: review and audit payroll dispute or adjustment forms
        - **Impact:** Introduces rigorous oversight of contentious payroll entries, ensuring each adjustment is justified and traceable.
    - Apply Payroll Changes: individual, team, or global adjustments to incentives, rates, or deductibles
        - **Impact:** Enables precise, scoped modifications without necessitating bulk spreadsheet edits, reducing the margin for human error.
    - Review Changes: audit trails and approval workflows
        - **Impact:** Embeds an immutable record of every alteration, promoting accountability and simplifying post-hoc forensic reviews.
    - Payroll Report: generate consolidated payroll summaries
        - **Impact:** Delivers comprehensive sums of payroll data across departments, expediting financial close processes and improving both managerial and company insight.
    - Distribute Payslips: secure generation and email delivery with retry logic

- **Impact:** Automates payslip dissemination, guaranteeing delivery even in the face of transient network failures and bolstering employee trust in timeliness.

2.3. Conceptual Framework

2.3.1. Use Case Diagram



2.3.2. Use Case Scenario

2.3.2.1. Basic Employee Interactions

An employee logs in, clocks in remotely, verifies on-site, checks today's attendance summary, views their salary estimate, and files an overtime request.

1. **Secure Login**
    a. Employee opens the app and enters EmployeeID, Password, 2FA Code.

b. System validates fields/format, checks password hash, verifies 2FA window, and applies progressive back-off on failures.
c. On success, old sessions for that ID are terminated, a new session is created, and the employee is redirected to their Branch Dashboard.

2. **Clock In On-Site**
    a. Employee taps Clock In: system confirms session, reads today's day type, prompts if it's a rest day/holiday/leave, then records off-site time and tags present or tardy.
    b. A QR code is displayed for 10 min.
    c. Upon arrival on-site, employee scans the QR: system validates it, updates status to "Onsite," recomputes tardy/present tag, and updates the attendance log.
3. **View Today's Attendance**
    a. Employee opens Attendance tab: system fetches today's clock-in/out and displays status ("Present," "Tardy").
4. **Salary Snapshot**
    a. Employee opens Salary Tracker: system retrieves their latest precomputed snapshot (hours, incentives, deductions, net pay) and shows "Estimate as of [timestamp]."
5. **File Overtime Ticket**
    a. Employee selects Overtime Request, fills fromDate/toDate/reason, and submits.
    b. System generates a TicketID, queues it to the team lead, and shows the employee their ticket's status and position in queue.

2.3.2.2. Team Management

A team lead reviews the team roster, adjusts a member's weekly schedule, approves a leave request, and edits a timesheet entry.

1. **View Roster & Availability**
    a. Lead logs in and opens Team Management: system validates session, fetches team members, today's DayType and attendance tags, and contact info.
    b. Roster UI shows each member's status (Present, OnLeave, Absent).
2. **Edit Weekly Schedule**
    a. Lead selects a member, chooses "Weekly Pattern," defines Mon-Fri 10:00-18:00, and clicks Save.
    b. System updates the member's schedule, logs the change, and notifies the member.
3. **Process Pending Tickets**

a. Lead clicks into Pending Requests, sees a team member's Leave Request.
b. Reviewing the form, the lead approves it; system updates the leave calendar and notifies the employee.

4. **Timesheet Correction**
    a. Lead opens the Timesheet Change ticket, sees requested clock times, and applies the correction to the attendance log.
    b. System marks the ticket Resolved, updates the log, and notifies the employee.

2.3.2.3. HR Management

An HR user onboards a new hire, updates an existing employee's branch, resolves an HR ticket, and processes a departure.

1. **Onboarding**
    a. HR clicks Add New Hire, enters name, birthdate, branch, position, pay rate, tax IDs, then submits.
    b. System validates inputs, assigns EmployeeID, creates the record, and adds it to the Onboard Queue.
    c. After asset setup, HR marks the hire "Complete," removing them from the queue.
2. **Record Update**
    a. HR searches an existing employee and changes their branch.
    b. System logs the change with timestamp and HRID and updates the employee record.
3. **HR Ticket Handling**
    a. An employee submits a "General Inquiry" about benefits.
    b. HR opens Tickets, reviews the form, replies in-system, marks it Resolved, and notifies the employee.
4. **Offboarding**
    a. HR clicks Process Departure for an exiting employee, enters departure date and clearance status, then submits.
    b. System adds them to the Offboard Queue; once final clearance is done, HR marks them Inactive and removes them from the queue.

2.3.2.4. Payroll Supervision

A payroll staff member applies a global tax-rate update, reviews an individual subsidy dispute, and then distributes final payslips.

1. **Apply Global Adjustment**
    a. Staff opens Payroll Adjustments and chooses Global Scope.
    b. In the Adjustment Form, selects withholding tax rate > replace > 12%, then submits.

c. System validates uniqueness, assigns dispute r equest ID, and sets status Pending.

2. **Approve Individual Dispute**
    a. Employee disputed “RiceSubsidy” via ticket; staff opens an Individual Adjustment Form, adds rice subsidy > add > ₱200, and submits.
    b. System loads recorded values, applies the change to that employee’s payroll totals, logs the audit, and regenerates their payslip.
3. **Distribute Payslips & Reports**
    a. After all approvals, staff clicks Distribute Payroll.
    b. System generates PDFs for each employee, emails them with retry logic for failures, and sends a collated report to the payroll team.
    c. A dashboard shows real-time status; on completion, the cycle is marked Distributed.
    d. An overall report summarising totals (gross, deductions, net by department or branch)

2.3.3. Logical View

2.3.3.1. System Users

RBAC Roles

System Admin	Manages system configuration, user roles, security policies, and full database access.
HR	All lead functions + onboard/offboard management, handles employee records, and other HR-related tickets.
Payroll Staff	Processes payroll, manages earnings/deductions, and generates disbursements.
Supervisor	All employee functions + manage team schedules, approve OT/leave, handle timesheet edits.
Employee	Clock in/out, view attendance, salary tracker, submit tickets.

RBAC Permissions Matrix

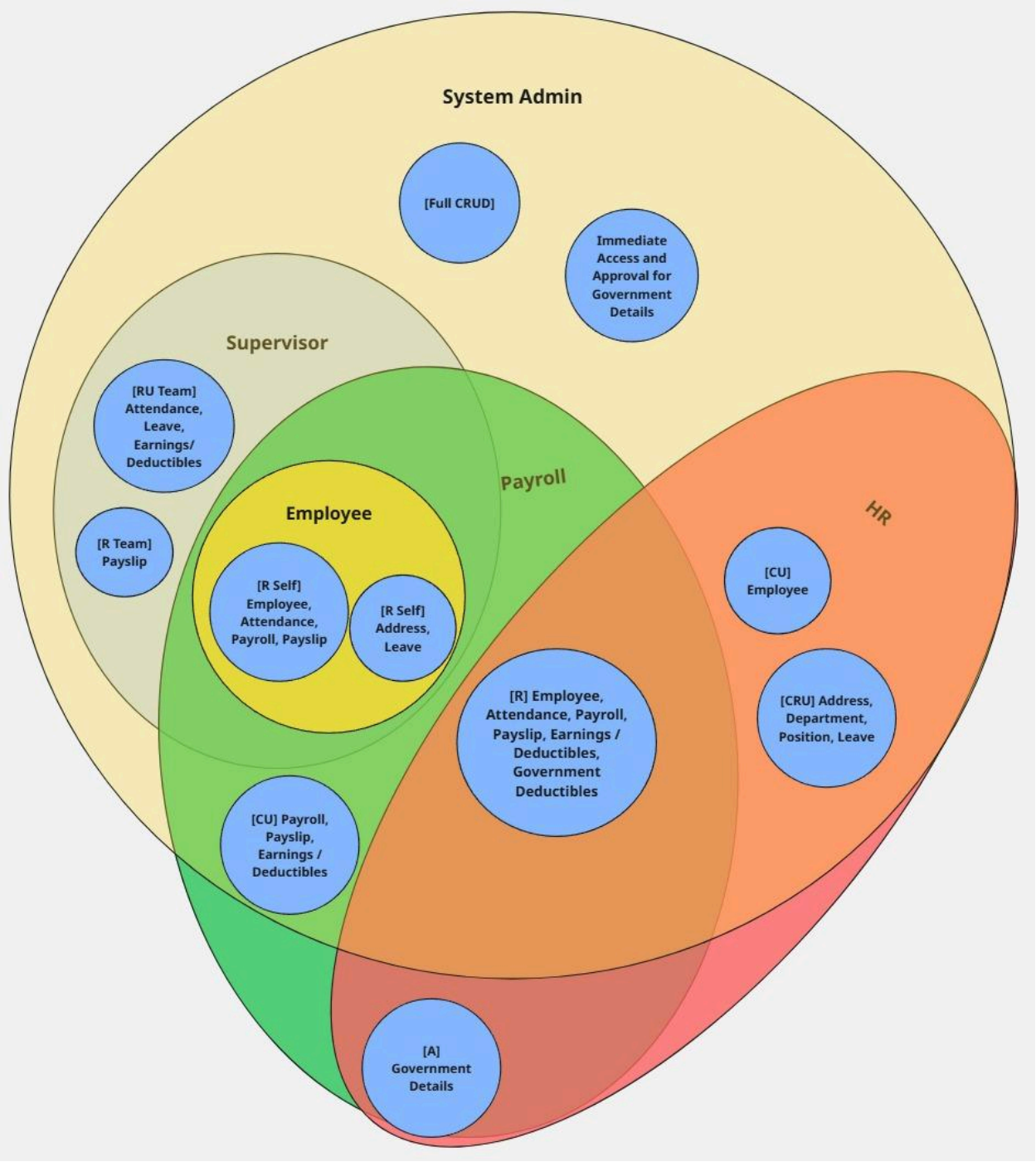
C: Create R: Read U: Update D: Delete A: Approval Req

Table/Entity	System Admin	HR	Payroll	Supervisor	Employee
Employee	C, R, U, D	C, R, U	R	R (Self)	R (Self)
Address	C, R, U, D	C, R, U	R (Self)	R (Self)	R (Self)
Department	C, R, U, D	C, R, U	-	-	-
Position	C, R, U, D	C, R, U	-	-	-

C: Create R: Read U: Update D: Delete A: Approval Req

Table/Entity	System Admin	HR	Payroll	Supervisor	Employee
Leave	C, R, U, D	C, R, U	R (Self)	R, U (Team)	R (Self)
Attendance	C, R, U, D	R	R	R, U (Team)	R (Self)
Payroll	C, R, U, D	R	C, R, U	R (Self)	R (Self)
Payslip	C, R, U, D	R	C, R, U	R (Team)	R (Self)
Earnings/Deductibles	C, R, U, D	R	C, R, U	R, U (Team)	-
Government Deductibles	C, R, U, D	R	R	-	-
Government Details	C, R, U, D	A	A	-	-
Tax Table/Rates	C, R, U, D	-	-	-	-

2.3.4. Euler's Diagram of the Payroll System Access



3. Conclusion

<Frame this section based on the outline:

- *Recap of Proposal Highlights - Start by outlining the proposal's primary features and critical elements. Remind the reader of the fundamental issue or opportunity, the suggested solution, and the advantages presented.*

- *Reiteration of Value - Reinforce the importance and significance of the suggested technology solution. Remind the reader how the solution solves the difficulties or possibilities mentioned in the introduction.*
- *Alignment with Business Goals - Highlight how the technology solution matches the business's overall goals and strategy. Restate how it contributes to the business's purpose and long-term strategy.*
- *Call to Action - Clearly express the intended action you want the reader to take. This could entail accepting the idea, scheduling more talks, or moving forward with execution. Encourage the reader to take the next step.>*

As we go forward, we look forward to working with you to shape your business's future. We are ready to begin conversations, establish the development plan, and drive the success we know this solution will bring.

Thank you for trusting us as we begin toward a more inventive, efficient, and profitable future.

4. Appendix

4.1. Definitions Acronyms and Abbreviations

<List important terms and acronyms throughout the document to ensure a common understanding among all readers. Use the table below.>.

Term	Definition

4.2. References

Reference No.	Reference Document
1	Business Requirement Specification Document
2	Requirements Analysis Document
3	System Design Document*